

22: PostgreSQL Projects

Real-World Projects from Beginner to Advanced

These projects bridge the gap between learning SQL syntax and building production-grade database systems. Each project is a complete, standalone application schema with realistic data, 20+ queries, stored procedures, triggers, and performance optimization guidance.

Project Index

Beginner Projects (Weeks 1-3)

#	Project	Core Skills	Estimated Time
01	Library Management System	Basic schema, CRUD, JOINs, aggregations, simple triggers	1 week
02	Student Management System	Recursive CTEs, weighted GPA, domains, many-to-many	1 week
03	Inventory System	Ledger pattern, materialized views, moving average cost, UPSERT	1 week

Intermediate Projects (Weeks 4-7)

#	Project	Core Skills	Estimated Time
04	E-Commerce Backend	JSONB, UUID, state machine, LATERAL joins, cohort analysis	2 weeks
05	CRM System	SCD history, soft deletes, funnel analysis, JSONB custom fields	2 weeks
06	Blogging Platform	FTS, partitioning, generated columns, polymorphic reactions	2 weeks

Advanced Projects (Weeks 8-13)

#	Project	Core Skills	Estimated Time
07	Banking System	Double-entry ledger, SERIALIZABLE, RLS, fraud detection, interest	3 weeks
08	Ride-Sharing Backend	PostGIS, KNN search, surge pricing, LISTEN/NOTIFY, geofencing	3 weeks
09	Event Streaming Platform	Append-only partitioning, funnel analysis, A/B testing, cohort retention	3 weeks

How to Use These Projects

Recommended Learning Path

```
BEGINNER
01 Library --> 02 Student --> 03 Inventory
    |               |
    v               v
INTERMEDIATE
04 E-Commerce --> 05 CRM --> 06 Blog
    |
    v
ADVANCED
07 Banking --> 08 Ride-Sharing --> 09 Event Platform
```

For Each Project

1. **Read the overview** — understand what you're building and why.
2. **Draw the ER diagram** on paper before looking at the ASCII version.
3. **Create the schema** from scratch — don't copy-paste. Type it.
4. **Load the sample data** and verify with SELECT queries.
5. **Run queries 1-10** and understand the output.
6. **Implement the stored functions** and test edge cases.
7. **Add the triggers** and verify they fire correctly.
8. **Run EXPLAIN ANALYZE** on the 3 most complex queries.
9. **Pick 2 Extension Challenges** and implement them.

Skills Matrix

Skill	Project(s)
Basic schema design	01, 02, 03
CHECK / UNIQUE / FK constraints	01, 02, 03, 04
PL/pgSQL functions	01, 02, 03, 04, 05, 06
Triggers	All projects
Window functions	01, 02, 03, 04, 05, 06, 07, 08, 09
Recursive CTEs	02, 05
JSONB	04, 05, 06, 08, 09
Full-text search	01, 06
Table partitioning	06, 07, 08, 09
PostGIS / geospatial	08
Row-Level Security	07

SERIALIZABLE transactions	07
Materialized views	03, 06, 09
Double-entry bookkeeping	07
A/B testing	09
Funnel analysis	09
Cohort retention	04, 09

Prerequisites

Before starting these projects, ensure you have completed:

- Chapters 01-06: SQL fundamentals through database design
- Chapter 07: Indexes (required for optimization sections)
- Chapter 09: Transactions (required for banking and inventory)
- Chapter 11: Advanced PostgreSQL features (required for projects 06-09)

Setup

```
-- Create a dedicated database for projects
CREATE DATABASE pg_projects;

-- Connect and create schemas
\c pg_projects

-- Each project uses its own schema
-- library, sms, inv, shop, crm, blog, bank, rides, analytics

-- Enable required extensions
CREATE EXTENSION IF NOT EXISTS pg_trgm;
CREATE EXTENSION IF NOT EXISTS pgcrypto;
CREATE EXTENSION IF NOT EXISTS postgis; -- Project 08 only
CREATE EXTENSION IF NOT EXISTS pg_stat_statements;
```

Evaluation Criteria

For each project, assess yourself against:

Criterion	Basic	Proficient	Expert
Schema design	Tables exist	Proper constraints	Optimized + normalized
Query correctness	Returns data	Correct results	Edge cases handled
Function quality	Works	Error handling	Performance aware

Index usage	No indexes	Basic indexes	Partial + covering
EXPLAIN usage	Never used	Used once	Used to optimize
Extension challenges	0 completed	2 completed	4+ completed

Connecting to the Interview Prep

Each project maps to common interview question categories:

Interview Topic	Covered In
"Design a database for X"	All projects
"Write a query for Y analytics"	04, 05, 09
"How would you handle concurrency?"	07, 08
"Explain your indexing strategy"	All projects
"How would you partition this table?"	06, 07, 08, 09
"What's your approach to data integrity?"	07 (banking)
"How do you handle soft deletes?"	05, 06
"Explain your caching strategy"	03, 06, 09